

Exercise Sheet 7

November 28th: OO-Principles II

Preparation:

1. You need to install IntelliJ's IDEA¹ and the following IntelliJ Plug-In:
 - (a) MetricsReloaded <https://github.com/BasLeijdekkers/MetricsReloaded> – but you can easily install it from within the IDE using the IntelliJ plugin manager.
 - (b) Once installed you can run several metrics including all from Robert C. Martin's book.
2. Get the sources for JUnit 3.8 (available in the Moodle course, see unit 4).
3. Get the sources for JHotDraw, see <http://sourceforge.net/projects/jhotdraw/files/latest/download> and <http://softarch.cis.strath.ac.uk/PLJHD/Patterns/JHDDomainOverview.html> or use the “old” 5.2. Version available in E-Learning.
4. Get the sources for Tomcat 8.54, see <https://tomcat.apache.org/download-80.cgi>

Exercise 1

A toy example!

1. Install MetricsReloaded
2. Run it on the HeadFirst examples (project).
3. Apart from “Uncle Bob” (Robert C. Martin)’s metrics try out some other metrics that are calculated by MetricsReloaded and think about them.

Exercise 2

Analyze JUnit!

1. Install and compile JUnit sources!
2. Run MetricsReloaded on the all JUnit packages and interpret the results! Are there any “bad” packages and why?
3. Run MetricsReloaded on the subset of all JUnit packages, namely `junit.framework`, `junit.runner`, and `junit.extensions`, and interpret the results! Any changes?
4. Run MetricsReloaded! What parameters are “questionable” or “interesting” for you and why?

Exercise 3

Analyze Tomcat!

1. Install and compile the Tomcat 8 from the source code sources!
2. Run MetricsReloaded on the all Tomcat `catalina` packages and interpret the results! Are there any “bad” packages and why (Hint: watch `util`!)?
3. Run MetricsReloaded! Are there any “bad” packages and why?

Exercise 4

Analyze JHotDraw!

1. Install and compile the JHotDraw sources!

¹all students of Frankfurt-UAS are entitled to downloading any JetBrains's product using their university's email account as authorisation

2. Run MetricsReloaded on the all JHotDraw **draw** packages and interpret the results! Are there any “bad” packages and why?
3. Run MetricsReloaded on the *all* JHotDraw packages and interpret the results! Any changes and why?
4. Run MetricsReloaded! Are there any “bad” packages and why?

Hints

- Consult the literature!
- You can work in pairs, if you want!
- If you want to learn a Java API, look into the java docs!
- Always use the same familiar IDE (suggestion JetBrains IntelliJ or Eclipse)!